BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA

Order Instituting Rulemaking to Modernize the Electric Grid for a High Distributed Energy Resources Future.	Rulemaking 21-06-017 (Filed June 24, 2021)

REPLY COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE ADMINISTRATIVE LAW JUDGE'S RULING PROVIDING TWO WORKING GROUP REPORTS AND DIRECTING RESPONSES TO QUESTIONS ON REPORTS

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission ("Commission"), the Vehicle-Grid Integration Council ("VGIC") hereby submits these reply comments on the *Administrative Law Judge's Ruling Providing Two Working Group Report And Directing Responses To Questions On Reports* ("Ruling"), filed May 29, 2024, as modified by the *Administrative Law Judges' Ruling Providing Uncorrupted Attachments,* filed June 3, 2024, and the *Email Ruling Partially Granting Extension of Time for Comments,* filed June 11, 2024.

I. INTRODUCTION.

In Opening Comments, VGIC recommended prioritizing the implementation of the Limited Load Profiles ("LLP"), and VGIC strongly supports The Public Advocates Office at the California Public Utilities Commission's ("Cal Advocates") recommendation to "adopt an accelerated policy development pathway to enable Limited Load Profiles."[1] Developing LLPs is crucial for optimizing the use of the distribution system and enhancing grid reliability and efficiency. VGIC urges the Commission to begin work on developing static LLPs that can be used

[1] Cal Advocates Opening Comments at 2.

as a bridge solution to allow electric vehicle ("EV") chargers to come online while grid upgrades are constructed. While the ongoing pilots by Pacific Gas and Electric ("PG&E") and Southern California Edison ("SCE") to implement flexible connection agreements are important efforts that should continue, the Commission should begin discussion on expanding and standardizing these efforts. Given the scale of the need for EV charging and the amount of distribution work that will be required, VGIC expects LLPs to be heavily leveraged to enable EVs to be adopted in California anywhere near the extent envisioned by CARB's Advanced Clean Cars II, Advanced Clean Fleets, and Advanced Clean Truck regulations.

In response to Opening Comments, VGIC would like to provide additional comments on the EV Business Case and cybersecurity questions. Overall, we recommend:

- Implement LLPs or import schedules that align with customer preferences and ensure adequate compensation mechanisms for responding to import limit modifications.

II. IMPLEMENT LLPS OR IMPORT SCHEDULES THAT ALIGN WITH CUSTOMER PREFERENCES AND ENSURE ADEQUATE COMPENSATION MECHANISMS FOR RESPONDING TO IMPORT LIMIT MODIFICATIONS.

VGIC appreciates the comments by parties on question 10 for the EV Business Case (Business Case E) on initial efforts to deploy chargers using flexible connection agreements. Overall, VGIC strongly supports the further development and rollout of PG&E's Flexible Service Connection pilot and SCE's Load Control Management Systems ("LCMS") pilot, as these will provide valuable insights for the Commission to further develop LLPs and other forms of firm and non-firm capacity agreements. Below, VGIC provides responses to party comments on questions 10c and 10d.

10c. What is the preferred temporal granularity for power import limiting and/or power export limiting schedules?

In Opening Comments, PG&E and SCE shared the existing temporal granularity for their flexible connection pilots. For PG&E, these are, "Static seasonal limits (e.g. 2 seasonal peak values); Static temporal limits (e.g. limits based on time of day); Static constant limit (same value 24/7/365); or, A combination of values."[2] For SCE, options are, "Flat seasonal limits (three limits per year); Flat seasonal limits with "on peak" and "off-peak values" (six limits); Year-round flat limit (one limit)."[3] While neither pilot has robust data yet to determine which schedules are preferred by customers or the utilities themselves for various reasons, PG&E does state that "PG&E typically provides a limit value that covers two times of day for two times of year."[4] This is in line with VGIC's response in Opening Comments to Question 6b, which recommends considering the 4-value structure described by PG&E and simpler structures more generally. VGIC notes that there may also be merit to a 288-value approach that mimics the adopted Limited Generation Profiles ("LGP") methodology. It is not yet reasonable to preclude any one approach, as customers – and utilities – may reveal a preference for a best-fit approach as the pilots continue.

PG&E also discusses how their flexible service connection pilot provides non-firm capacity in the form of "day-ahead hourly limits, which may be adjusted in real-time under emergency scenarios."[5] VGIC looks forward to learning from the PG&E pilot how customers leveraged non-firm capacity and utilized the day-ahead schedule. Given the lack of experience with import limiting schedules, it is unclear whether this type of hourly schedule provided on a day-ahead basis is optimal or preferred by customers. **Generally, customers' preference to**

[2] PG&E Opening Comments at 10.
[3] SCE Opening Comments Attachment at 16.
[4] PG&E Opening Comments at 8.
[5] PG&E Opening Comments at 11.

leverage non-firm capacity with a particular schedule and notice timeline will depend on (a) the costs of implementing the capability to respond and (b) the price signals and compensation provided for responding. In some cases, new load customers may be interested in providing the utility with non-firm import capacity in exchange for a reduced energization timeline in lieu of compensation or for reduced compensation. For the implementation of Business Case B, where *firm* import limits are modified, VGIC reiterates that adequate compensation must be provided to respond to these signals and that the desired temporal granularity for responding to these *firm* import modifications will depend wholly or primarily on the compensation mechanisms implemented for DER to respond to utility import limit requests.

10d. What communication pathways or protocols, if any, are required for the DSOs to communicate with and send commands to the EVSE at EV charging sites?

In opening comments, SCE notes that DSOs should implement IEEE 2030.5 as the communication protocol for EV use cases, in line with other DER use cases.[6] Enphase recommends that the utilities, as DSOs, should incorporate IEEE 2030.5 into their backend communication systems now instead of waiting for the development of full DER management systems ("DERMS").[7]

VGIC would like to emphasize that DSO communication may not be directly to the individual EVSE or EV charging sites but rather to a third-party Charge Point Operator ("CPO") or aggregator managing energy usage at EV charging sites. This was discussed by SCE in Opening Comments as shown in Figure 2. The third party may receive the signals communicated by the DSO, for example, from a DERMS or other IEEE 2030.5 signaling system, and then use best-fit

[6] SCE Opening Comments Attachment at 16.
[7] Enphase Opening Comments at 8-9.

protocols to communicate to the EVSE or EV charging site. These communication protocols may include OCPP (e.g., OCPP 1.6j, OCPP 2.0.1, or in-development OCPP 2.1), DNP3, and proprietary messaging applications. SCE has also noted OpenADR as a possible protocol.

III. CONCLUSION.

VGIC appreciates the opportunity to provide these reply comments on the Working Group Reports. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

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Date: July 22, 2024